College Physics II
Lecture summaries

DRAFT

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Abstract

Disclaimer: These notes *summarize* the key concepts and formulae discussed during lecture. They should be supplemented by any standard algebra-based physics textbook on these topics (e.g., by Rex and Wolfson) or internet resource to fill in any gaps. Please send any comments, criticisms, suggestions to: `joseph.d.romano@gmail.com`.

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1 Electric charge, electric force

1.1 Electric charge

- Properties of electric charge:

  (i) Two types: *positive* and *negative* (+ and −), whose effects tend to cancel. Opposite charges attract, like charges repel.

  (ii) Total electric charge is *conserved* (cannot be created or destroyed, only transferred from one object to another).

  (iii) Electric charge is *quantized* (integer multiples of the charge of an proton $e = 1.602 \times 10^{-19}$ C; an electron has charge $-e$).

- Materials:

  (i) *Conductors*: one or more electrons per atom are able to move freely through the material. Examples: aluminum, copper, gold, ... (metals).

  (ii) *Insulators*: electrons are bound to atomic nuclei. Examples: rubber, wood, paper, ...

- Some demos:

  1) A plastic rod rubbed with fur is negatively charged; a glass rod rubbed with silk is positively charged. Show that opposite charges attract, like charges repel.

  2) Bits of uncharged paper will be attracted to both negative and positive rods due to charge separation within the atoms / molecules in the paper (an insulator).

  3) A charged rubber ballon sticks to the white board for the same reason as (2).

1.2 Electric force

- *Coulomb's law*: The magnitude of the electric force between two point charges q_1 and q_2 separated by a distance r is

$$F = \frac{k|q_1||q_2|}{r^2} \quad \text{where} \quad k = 8.99 \times 10^9 \, \frac{\mathrm{N \cdot m^2}}{\mathrm{C^2}} \tag{1}$$

- The force is attractive (directed towards one another) if the two charges have opposite signs; the force is repulsive (directed away from one another) if the two charges have the same signs.

- One sometimes writes

$$k = \frac{1}{4\pi\epsilon_0}\,, \quad \text{where} \quad \epsilon_0 = 8.85 \times 10^{-12} \, \frac{\mathrm{C^2}}{\mathrm{N \cdot m^2}} \tag{2}$$

  where ϵ_0 is the so-called *permitivity* of free space (vacuum).

- The electric force on point charge Q due to a set of point charges q_1, q_2, ..., q_N is given by the vector sum of the forces

$$\vec{F}_{Q,\mathrm{net}} = \vec{F}_{1Q} + \vec{F}_{2Q} + \cdots + \vec{F}_{NQ} \tag{3}$$

  where $\vec{F}_{1Q}$ is the electric force that point charge q_1 exerts on Q, etc. This is called the *superposition principle*.